PLURALCODE ACADEMY

Data Analysis Programme — February Cohort

CAPSTONE PROJECT

Exploratory Data Analysis

Cohort:	Data Analysis – February Cohort
Track:	Data Analysis with Python
Student Name:	________________________________
Submission Date:	________________________________

1. Project Overview

This capstone project brings together everything you have learned across Sessions 1 to 10 of the Data Analysis programme — from Python fundamentals through to Exploratory Data Analysis (EDA). You will act as a Junior Data Analyst asked to investigate a retail company's customer and sales data, clean it, explore it, visualize it, and present findings that a non-technical manager could act on.

You will build a single, well-organised Python script (or Jupyter notebook) plus a short written report. The project is designed to take a motivated student 6–10 hours spread across one week.

2. Learning Objectives Assessed

- Write clean, well-commented Python programs using variables, data types, and string formatting (Session 1)
- Use conditionals, loops, and functions to process data (Sessions 2–4)
- Create and manipulate lists, dictionaries, and NumPy arrays (Sessions 4–5)
- Load, clean, and explore data using Pandas Series and DataFrames (Session 6)
- Perform GroupBy, merge/join, and pivot-table operations (Session 7)
- Produce clear, labelled data visualizations with Matplotlib and Seaborn, including multiple charts on one figure grid (Session 8)
- Apply descriptive statistics, correlation, and hypothesis testing (Session 9)
- Carry out a structured Exploratory Data Analysis workflow and report insights (Session 10)

3. The Dataset

You will generate your own synthetic dataset so that every student's numbers are unique. Copy the code block below into your script exactly as written — do not change the seed — then complete the tasks that follow using this DataFrame.

```
import pandas as pd
import numpy as np

np.random.seed(42)          # keep this seed — do not change it
n = 500

df = pd.DataFrame({
    'CustomerID'   : range(1, n + 1),
    'Age'          : np.random.randint(18, 70, n),
    'Region'       : np.random.choice(['North', 'South', 'East', 'West'], n),
    'Income'       : np.random.randint(20000, 150000, n),
    'Purchases'    : np.random.poisson(5, n),
    'Member_Years' : np.random.randint(0, 10, n),
    'Satisfaction' : np.random.randint(1, 11, n)
})
```

```
# Introduce a few realistic problems for you to find and fix:
df.loc[np.random.choice(df.index, 15, replace=False), 'Income'] = np.nan
df = pd.concat([df, df.sample(5, random_state=1)], ignore_index=True)
```

This mirrors the customer dataset used in Session 10, but with missing values and duplicate rows added deliberately — part of your job is to find and handle them.

4. Project Tasks

Part A — Python Foundations Warm-Up (10 marks)

- Write a function calculate_clv(row) that estimates each customer's rough lifetime value as Purchases × Member_Years × 15, using conditionals and arithmetic from Sessions 1–4.
- Apply the function to create a new CLV column using a loop or .apply().

Part B — Data Cleaning (15 marks)

- Check for and report missing values and duplicate rows.
- Handle the missing Income values using an appropriate method (e.g. median imputation) and justify your choice in a comment.
- Remove duplicate rows and confirm the new shape of the DataFrame.

Part C — Exploration & Aggregation (20 marks)

- Use describe() to summarise the numeric columns.
- Use groupby() to find average Income and average Satisfaction per Region.
- Create a pivot table showing average Purchases by Region and an Age band (Under 30 / 30–50 / Over 50) that you engineer using a function or pd.cut().

Part D — Visualization: Combined Grid (25 marks)

This is the core visualization requirement for the capstone: you must plot two graphs on a single grid (one figure, two subplots) using plt.subplots(), not two separate plt.figure() calls.

- Left subplot: a bar chart of average Income by Region.
- Right subplot: a scatter plot of Age (x-axis) vs Purchases (y-axis), coloured or styled by Region.
- Both subplots must share one figure, have titles, axis labels, and a legend where relevant, and be saved as one image file (e.g. capstone_dashboard.png).

```
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Left subplot: bar chart
region_income = df.groupby('Region')['Income'].mean()
axes[0].bar(region_income.index, region_income.values, color='#1F3864')
axes[0].set_title('Average Income by Region')
```

```
axes[0].set_xlabel('Region')
axes[0].set_ylabel('Average Income ($)')

# Right subplot: scatter plot
axes[1].scatter(df['Age'], df['Purchases'], alpha=0.6, c='#B08D57')
axes[1].set_title('Age vs. Purchases')
axes[1].set_xlabel('Age')
axes[1].set_ylabel('Number of Purchases')

plt.tight_layout()
plt.savefig('capstone_dashboard.png', dpi=150)
plt.show()
```

You are encouraged to go further — e.g. a 2×2 grid with four related charts — for bonus marks, as long as the two required charts above are included.

Part E — Statistics (15 marks)

- Calculate the correlation between Income and Satisfaction, and between Member_Years and CLV.
- Split customers into two groups (e.g. Satisfaction ≥ 7 vs Satisfaction < 7) and run an independent-samples t-test comparing their Income using scipy.stats.ttest_ind.
- State in a comment whether the difference is statistically significant at the 0.05 level and what that means in plain English.

Part F — Insights & Recommendations (15 marks)

- Write a short written report (300–500 words, in a separate Word or PDF document, or as Markdown cells if using a notebook) summarising: your key findings, the two visualizations and what they show, your statistical result, and two concrete, actionable recommendations for the retail company.

5. Grading Rubric

Criterion	Marks	What we're looking for
Part A – Python Foundations	10	Correct, readable function and application logic
Part B – Data Cleaning	15	Missing values and duplicates correctly identified and handled
Part C – Exploration & Aggregation	20	Correct use of describe(), groupby(), and pivot_table()
Part D – Combined Grid Visualization	25	One figure, two clearly labelled subplots, saved correctly
Part E – Statistics	15	Correct correlation and t-test, sensible interpretation
Part F – Insights & Recommendations	15	Clear, specific, and data-backed recommendations
Total	100	

6. Submission Guidelines

- Submit one .py file or .ipynb notebook, plus the saved capstone_dashboard.png, plus your written report.
- Name your files: LastName_FirstName_Capstone.py / .ipynb
- Code must run top-to-bottom without errors. Comment your code so a classmate could follow your thinking.
- Deadline: to be confirmed by your facilitator — typically one week after Session 10.
- Submit through the class portal or channel shared by your facilitator.

7. Academic Integrity

You may discuss approaches with classmates, but the code, analysis, and written report you submit must be your own work. You are welcome to reuse patterns from the Session 1–10 course materials, but copy-pasted, unexplained code will lose marks under Part A and Part F.

Need Help?

Revisit the relevant session file if you get stuck: Sessions 1–4 for Python basics, Session 5 for NumPy, Session 6–7 for Pandas, Session 8 for visualization, and Sessions 9–10 for statistics and EDA. Bring specific error messages (not just “it doesn't work”) to office hours or the class channel.